Math 1232 Summer 2024
Mastery Quiz 1
Due Thursday, July 3

This week's mastery quiz has two topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn this quiz in class on Thursday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 1: Transcendental Functions
- Secondary Topic 1: Invertible Functions

Name: Solutions

Name: Solutions

Major Topic 1: Transcendental Functions

(a) Compute $\displaystyle\int_0^{\ln(5)} e^x\sqrt{1+3e^x}\,dx$

Solution. Set $u = 1+3e^x$, so $du = 3e^x\,dx$, and we see that $u(0) = 4$ and $u(\ln(5)) = 16$.

$$\begin{aligned}\int_0^{\ln(5)} e^x\sqrt{1+3e^x}\,dx &= \int_4^{16} \frac{1}{3}\sqrt{u}\,du \\ &= \frac{2}{9}u^{3/2}\Big|_4^{16} = \frac{2}{9}\cdot 64 - \frac{2}{9}\cdot 8 = \frac{112}{9}.\end{aligned}$$

(b) Find an equation for the tangent line to the curve $y = \ln(2x^2-3x-1)$ at the point $(2,0)$.

Solution. We have $y' = \frac{1}{2x^2-3x-1}(4x-3)$ and thus $y'(2) = \frac{5}{1} = 5$. So the equation of the tangent line is

$$y - 0 = 5(x-2).$$

(c) Compute $\displaystyle\int \frac{\sin(2t)+\cos(2t)}{\sin(2t)-\cos(2t)}\,dt.$

Solution. We can take $u = \sin(2t)-\cos(2t)$. Then $du = (2\cos(2t)+2\sin(2t))dt$, and we get

$$\begin{aligned}\int \frac{\sin(2t)+\cos(2t)}{\sin(2t)-\cos(2t)}\,dt &= \int \frac{\sin(2t)+\cos(2t)}{u}\,\frac{du}{2\cos(2t)+2\sin(2t)} \\ &= \int \frac{1}{2}u\,du = \frac{1}{2}\ln|u| + C \\ &= \frac{1}{2}\ln|\sin(2t)-\cos(2t)| + C.\end{aligned}$$

Name: Solutions

Secondary Topic 1: Invertible Functions

(a) Is $f(x) = x^2 + x$ invertible or not? Justify your answer.

Solution. We have $f(-1) = f(0) = 0$ so this function is not one-to-one, and thus not invertible.

(b) Give an exact solution (with no decimals) for the equation $\ln(2x + 5) = 3$.

Solution.

$$\begin{aligned} \ln(2x+5) &= 3 \\ 2x + 5 &= e^3 \\ 2x &= e^3 - 5 \\ x &= \frac{e^3 - 5}{2}. \end{aligned}$$

(c) Let $h(x) = \sqrt{x^3 + x + 6}$. Compute $(h^{-1})'(4)$.

Solution. By the Inverse Function Theorem, we know that

$$(h^{-1})'(4) = \frac{1}{h'(h^{-1}(4))}.$$

Guess and check shows that $h(2) = 4$ so $h^{-1}(4) = 2$. And we know that

$$h'(x) = \frac{1}{2}(x^3 + x + 6)^{-1/2}(3x^2 + 1)$$

and thus

$$h'(2) = \frac{1}{2}(16)^{-1/2}(13) = \frac{13}{8}.$$

Thus

$$(h^{-1})'(4) = \frac{1}{13/8} = \frac{8}{13}.$$